

```
!pip install gensim nltk
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
27.9/27.9 MB 55.3 MB/s eta 0:00:00

Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
import nltk
nltk.download('punkt')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
True
```

```
from gensim.models import Word2Vec
```

```
!pip install gensim
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
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Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
```

```
# Import required library
from gensim.models import Word2Vec

# Step 1: Create a mini corpus (10-15 sentences)
corpus = [
    "The student studies machine learning",
    "The teacher explains the lesson clearly",
    "Students learn programming and data science",
    "The computer processes data quickly",
    "Artificial intelligence improves modern technology",
    "The government supports education and research",
    "Money is important for economic development",
    "The university provides quality education",
    "Data science uses statistics and programming",
    "Technology changes the future of education",
    "The student uses a computer for studying",
    "Researchers develop new artificial intelligence models"
]

# Step 2: Preprocess sentences (tokenization using split)
tokenized_sentences = []

for sentence in corpus:
    tokens = sentence.lower().split()
    tokenized_sentences.append(tokens)

# Step 3: Train Word2Vec model using Skip-gram
model = Word2Vec(
    sentences=tokenized_sentences,
    vector_size=50,
    window=3,
    min_count=1,
    workers=4,
    sg=1 # Skip-gram model
)

# Step 4: Generate vocabulary
vocab = list(model.wv.index_to_key)

print("\nVocabulary Words:")
print(vocab)

# Step 5: Query words
query_words = ["student", "computer", "education"]
```

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# Step 6: Find top 5 similar words
for word in query_words:

    print("\nTop 5 similar words for:", word)

    similar_words = model.wv.most_similar(word, topn=5)

    for w, score in similar_words:
        print(w, ":", round(score, 3))
```

Vocabulary Words:

['the', 'education', 'data', 'and', 'uses', 'for', 'technology', 'intelligence', 'artificial', 'computer', 'science', 'progr

Top 5 similar words for: student

government : 0.191

and : 0.19

new : 0.188

supports : 0.185

learn : 0.175

Top 5 similar words for: computer

important : 0.265

teacher : 0.255

explains : 0.236

for : 0.225

university : 0.2

Top 5 similar words for: education

money : 0.376

researchers : 0.23

important : 0.221

university : 0.219

government : 0.195